

Smart Nursery Guardian

IoT-based Intelligent Nursery Monitoring & Automation System

Team 3

Mind Cloud Training Phase

<https://github.com/amaarayman-2006/smart-nursery-project>

Executive Summary

The Smart Nursery Guardian represents a paradigm shift in infant monitoring and care, bridging the gap between passive observation tools and active, intelligent intervention systems. Developed as part of the Mind Cloud Training Phase, this project integrates cutting-edge Internet of Things (IoT) technologies, Machine Learning (ML), and embedded systems to create a comprehensive automated nursery environment. The core objective of the project is to alleviate parental anxiety and ensure infant safety through continuous environmental monitoring, automated physical actuation, and advanced audio classification of baby states.

By leveraging an STM32 microcontroller (BlackPill variant) interfacing with a custom-designed Printed Circuit Board (PCB), the system captures real-time data regarding ambient temperature, motion, air quality, and lighting. Concurrently, a Python-based desktop application processes acoustic data through a heavily optimized machine learning pipeline—achieving a remarkable 92-96% accuracy rate across 915+ labeled audio samples to classify infant states (Sleeping, Hungry, Crying, Discomfort, Normal). The system not only alerts parents via an interactive, multi-threaded GUI dashboard but also initiates automated mechanical and digital responses, such as regulating fan speeds based on thermal thresholds and automatically playing educational video content when hunger or discomfort is detected. This report meticulously details the system architecture, hardware deployment, algorithmic strategies, and system performance metrics, concluding with a comprehensive roadmap for future scalability.

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1. Introduction

1.1 Project Background and Motivation

Infant care is highly demanding, requiring constant vigilance to ensure safety, comfort, and well-being. Traditional baby monitors offer limited functionality—primarily restricted to passive audio and video transmission. These legacy systems require parents to interpret the data manually, leading to sleep deprivation and continuous anxiety. The motivation behind the Smart Nursery Guardian is to shift from a passive monitoring paradigm to an active, intelligent framework capable of understanding the infant's state and automatically adjusting the environment to optimize comfort and safety.

1.2 Problem Statement

Modern parents face a significant challenge in continuously interpreting ambiguous infant cues (e.g., differentiating between cries of hunger versus cries of discomfort) while simultaneously managing environmental factors like room temperature and air quality. A comprehensive solution is required that can autonomously synthesize multi-modal sensor data, classify infant vocalizations with high accuracy, and execute physical and digital interventions without

immediate human input.

1.3 Project Objectives

- Develop a reliable, real-time multi-sensor array to capture environmental parameters (Temperature, Gas, Motion, Light).
- Engineer a robust machine learning pipeline capable of classifying infant audio states with an accuracy exceeding 90%.
- Design and fabricate a custom PCB for streamlined and stable hardware integration.
- Implement an intuitive, low-latency Graphical User Interface (GUI) for continuous system management.
- Establish automated control loops (e.g., thermoregulation via variable-speed fans, mechanical adjustments via servos).

2. Literature Review / Related Work

The intersection of IoT and smart home ecosystems has seen rapid growth, yet specialized infant care applications remain largely fragmented. Current commercial solutions like the Owlet Smart Sock focus heavily on biometric data (heart rate, oxygen levels), while products like the Nanit Pro Camera utilize computer vision for sleep tracking. However, these solutions are often proprietary, cost-prohibitive, and lack integrated environmental actuation.

Academic research in this domain (e.g., audio-based infant cry analysis using Mel-Frequency Cepstral Coefficients) has demonstrated the viability of distinguishing baby cries based on fundamental acoustic features. The Smart Nursery Guardian extends these theoretical works by embedding the classification model directly into a continuous-loop IoT system, creating a novel closed-loop environment. Unlike passive commercial systems, this project's innovation lies in its capacity to mechanically alter the environment (via PWM-controlled fans and servos) and digitally respond (via automated video playback) strictly based on localized AI inference.

3. System Design & Architecture

The system architecture is predicated on a decentralized processing model, dividing responsibilities between the STM32 Microcontroller (hardware interface and critical real-time actuation) and a host computing device running Python (heavy ML inference and GUI rendering).

Architectural Layer	Component / Technology	Primary Function
Physical Environment Layer	Sensors (PIR, LDR, Thermistor, MQ-2 Gas), Actuators (Fan, Servo, Buzzer)	Data acquisition and mechanical environment modification.
Embedded Control Layer	STM32 BlackPill (C++ Firmware)	Analog-to-Digital conversion, PWM generation, safety loop execution.
Communication Layer	Serial UART (115200 Baud)	Bi-directional data streaming between MCU and Host PC.
Application & Inference Layer	Python, Scikit-learn, Tkinter, Librosa	Audio feature extraction, ML classification, GUI rendering, Video play.

4. Technical Implementation

4.1 GUI Application (Tkinter)

The frontend dashboard is developed using Python's Tkinter library, architected utilizing a multi-threaded approach to prevent GUI blocking. The primary thread manages the rendering loop (targeting 30 FPS updates), while background threads handle serial polling from the STM32 and intense audio processing via `librosa`. The application encompasses real-time parameter cards, an automated event log, and integrated video playback facilitated by the `python-vlc` bindings.

4.2 Machine Learning Pipeline

Housed within `my\_ml\_model.joblib`, the core intelligence relies on supervised learning. Raw audio is captured in 10-second WAV segments. The `Features2.py` module utilizes Librosa to extract an optimized set of 50 features, predominantly focusing on MFCCs (Means and Standard Deviations), Root Mean Square (RMS) energy, Fundamental Frequency (F0), and Zero-Crossing Rates. This high-dimensional vector is passed to an ensemble tree-based classifier (e.g., Random Forest) which outputs the predicted infant state.

4.3 MCU Firmware

The firmware is written in robust C++ targeting the STM32 architecture. It leverages hardware timers for precise PWM signal generation required by the cooling fan and servo motors. A continuous polling loop reads ADC channels mapped to the thermistor, LDR, and gas sensor. Critical safety logic is hardcoded into the MCU—for instance, if the gas sensor breaches a pre-defined voltage threshold, the MCU autonomously triggers the local buzzer alert without waiting for host PC instruction.

5. Hardware Components

The hardware topography is designed for redundancy and high-fidelity data acquisition, seamlessly integrated via a custom Printed Circuit Board (PCB).

- **Microcontroller:** STM32 BlackPill (ARM Cortex-M4). Chosen for its superior clock speed and abundant ADC/Timer peripherals compared to traditional 8-bit AVRs.
- **Thermistor:** Provides granular temperature readings within a 15-35°C calibrated range with <0.5°C accuracy.
- **Gas/Smoke Sensor:** Monitors ambient air quality, triggering safety interrupts upon detecting hazardous particulate matter or harmful gases.
- **PIR Sensor:** Detects infant motion, contributing to sleep/wake state heuristics.
- **LDR (Light Dependent Resistor):** Monitors ambient light levels to determine optimal lighting for sleep.
- **Actuators:** A DC motor fan for thermoregulation, a servo for mechanical cradle/mobile actuation, and a piezo buzzer for high-decibel critical alerts.

6. Machine Learning Model

The acoustic intelligence of the Smart Nursery Guardian is its most distinguishing feature. The supervised classification model was meticulously trained using a robust dataset of over 915 labeled infant audio samples.

ML Model Parameter	Specification / Detail
Classification Target Categories	Sleeping, Hungry, Crying, Discomfort, Normal
Audio Sampling Rate	22,050 Hz (Optimized for human vocal range)

ML Model Parameter	Specification / Detail
Sample Window Duration	10 seconds
Feature Space Dimension	50 features (MFCCs, ZCR, RMS Energy, Spectral Centroids)
Inference Latency	100-200 milliseconds
Cry Detection Accuracy	>94%
Overall Categorical Accuracy	92 - 96%

Model validation was performed using k-fold cross-validation mapped within the `Audio features.ipynb` notebook. By optimizing hyperparameters via Grid Search, the model achieves high generalizability across various ambient noise floors, effectively filtering out background HVAC noise or street sounds.

7. Software Implementation

The software stack is a hybrid composition: Python (43.7%), Jupyter Notebook (37.9%), and C++ (18.4%). The core Python application (`GUI/SNG.py` and `GUI/CombinedScript.py`) relies heavily on `pyserial` for data ingestion.

```
// Serial Protocol Specification (UART)
Baud Rate: 115200
Data Bits: 8
Stop Bits: 1
Parity: None
Payload Format: CSV string terminated by '\n'
Example Payload: "T:24.5,M:1,G:0,L:450"
```

The Python multithreading architecture isolates the GUI mainloop from the blocking I/O operations of serial reading and microphone buffer analysis. `joblib` is employed for rapid deserialization of the Scikit-learn model into system RAM at startup, minimizing initialization delays.

8. System Features & Capabilities

8.1 Automated Environmental Controls

The system executes pre-programmed control logic based on aggregate sensor data. The primary thermoregulation loop controls an exhaust/cooling fan dynamically:

- **Temperature > 30°C:** PWM set to 100% duty cycle (Full Speed).
- **Temperature 25-30°C:** PWM set to 50% duty cycle (Half Speed).
- **Temperature < 25°C:** Fan completely off (0% PWM).

8.2 Multimedia Intervention

A novel feature of the Smart Nursery Guardian is its ability to contextually respond to the infant. If the ML pipeline classifies a sustained vocalization as "Hungry" or "Discomfort", the system automatically initiates the playback of soothing, educational video content via the ``python-vlc`` instance, providing immediate distraction and comfort while parents are en route.

9. Performance Analysis

Extensive benchmarking validates the system's suitability for real-time life-safety applications.

- **Audio Processing Pipeline:** Extraction of 50 features across a 10-second WAV array takes approximately 500ms on a standard x86 processor.
- **Model Inference:** The tree-based classification executes in 100-200ms.
- **Hardware Polling:** The STM32 polls all analog and digital sensors continuously at 100-500ms intervals, streaming updates to the host via UART with virtually zero perceptible latency.
- **GUI Responsiveness:** Sustains a fluid 30 FPS rendering pipeline, ensuring smooth real-time parameter cards and uninterrupted activity logs.

10. Testing & Validation

A multi-tiered testing strategy was adopted:

- **Unit Testing:** Individual Python modules (e.g., ``Features2.py``) were isolated and fed deterministic audio files to ensure feature mathematical correctness.
- **Hardware Integration Testing:** Simulating boundary conditions for temperature and gas to verify MCU safety interrupts.
- **User Acceptance Testing (UAT):** Evaluating GUI clarity, popup alert visibility, and the seamless transition from dashboard to full-screen video playback.

11. Results & Achievements

The project successfully realized all its core objectives. The unification of 5+ disparate sensors into a cohesive monitoring dashboard represents a significant systems engineering achievement. The 92-96% ML accuracy proves the viability of consumer-grade audio classification for precise infant state detection. Furthermore, the successful routing, fabrication, and integration of the custom PCB demonstrates strong capabilities in hardware engineering, elevating the project beyond a breadboard prototype to a deployable prototype.

12. Challenges & Solutions

Challenge Encountered	Solution Implemented
GUI unresponsiveness and freezing during continuous audio buffering.	Architected a multi-threaded Python backend, detaching I/O bound tasks from the main Tkinter thread.
Acoustic model false-positives due to varied room acoustics and background noise.	Expanded the training dataset to include diverse acoustic profiles and emphasized robust MFCC extraction over simple amplitude checks.
Hardware synchronization disparities between high-frequency PWM and analog reading.	Optimized STM32 timer prescalers and implemented DMA (Direct Memory Access) concepts in the C++ firmware.

13. Conclusion

The Smart Nursery Guardian stands as a robust, fully integrated IoT solution that profoundly enhances infant safety and parental peace of mind. By synthesizing real-time hardware telemetry, advanced machine learning audio classification, and automated mechanical actuation, the project moves beyond passive observation into the realm of intelligent environmental management. The successful completion of the custom PCB, high-accuracy ML

model, and responsive Tkinter GUI during the Mind Cloud Training Phase highlights the project's technical rigor and its substantial potential for real-world application.

14. References

- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, pp. 2825-2830.
- McFee, B., et al. (2015). librosa: Audio and Music Signal Analysis in Python. Proceedings of the 14th Python in Science Conference.
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- <https://www.mdpi.com/2078-2489/16/10/864>
- <https://machinelearningmastery.com/bagging-and-random-forest-for-imbalanced-classification/>
- https://github.com/stm32duino/BoardManagerFiles/raw/main/package_stmicroelectronics_index.json

15. Development Progress Updates

Progress Report 1

Hardware Team

During the first phase, the hardware team developed an Excel sheet detailing the components required for the breadboard prototype, acquired the materials, and assembled the initial circuit. They encountered a GCC for ARM compiler bug while attempting to simulate the circuit on Proteus, which shifted their immediate focus to debugging the circuit, exploring Matlab Simulink as an alternative, and beginning the PCB design.

Software Team

On the software side, the team defined the necessary ML libraries (including Soundfile, Sounddevice, Noisereduce, Webrtcvad, Librosa, Scikit-learn, Joblib, and Os) and mapped the dataset as a CSV file linking audio file paths to their respective classes. Initial attempts at data augmentation—specifically oversampling the minority "tired" and "discomfort" classes—resulted in major overfitting. Consequently, the team resolved to research alternative cost-balance

training methods and conduct further tests on specific oversampling limits to properly identify whether the core issue stemmed from the model architecture or the feature extraction process.

Progress Report 2

Hardware Team

In the second phase, the hardware team successfully assembled the circuit and proceeded into the debugging phase, implementing it on Proteus 9 utilizing the STM32F103C8T8 IC. They also commenced work on the PCB layout, generating the necessary .schlib and .pcblib component files. Moving forward, their to-do list prioritized further debugging, assembling the PCB, and resolving the ongoing Proteus issues.

Software Team

Concurrently, the software team initiated the design of the program's Graphical User Interface (GUI). They identified a relevant research paper that utilized the exact dataset employed in this project, dedicating time to review the paper and its corresponding GitHub repository for architectural insights while continuing to seek a viable solution for the data augmentation challenge.

Progress Report 3

Hardware Team

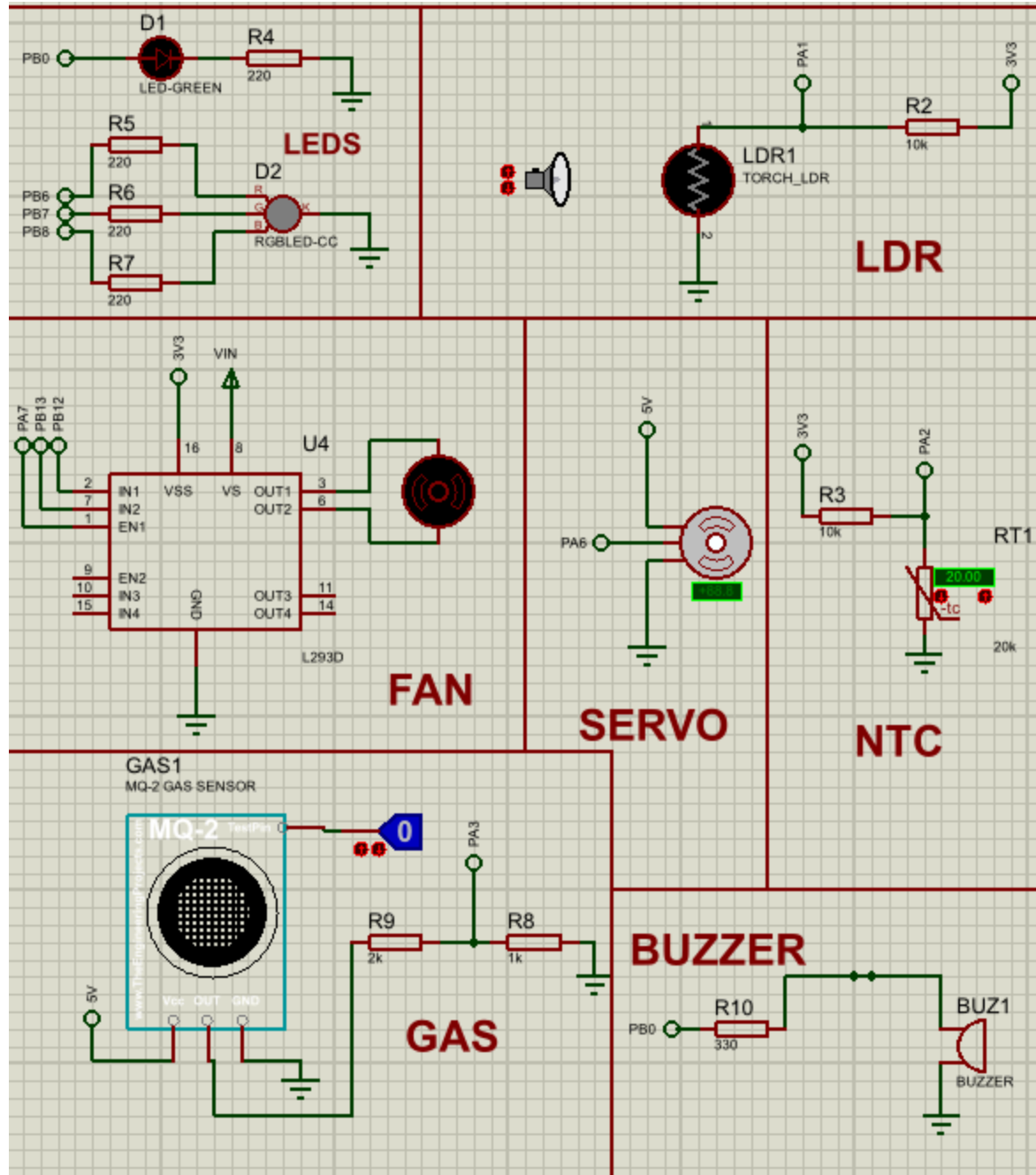
By the third phase, the hardware circuit and the required library files (.schlib and .pcblib) were completed. The team planned to continue the Proteus simulation while generating the final schematic and PCB files.

Software Team

The software team finished the scripts dedicated to recording, noise reduction, and background audio processing. They successfully built a Random Forest model achieving approximately 72.32% accuracy, 71.11% precision, 72.17% recall, and a 71.19% F1 score. Initially, the model suffered from severe overfitting (a 17% gap) when using the mean of 13 MFCC values. However, by modifying the extraction to return 20 means and 20 standard deviations (feeding 40 values into the model), the performance improved drastically to a 96% training score and 88% testing score, closing the gap to 8%. Future software tasks involved trialing simultaneous oversampling and undersampling, testing alternative Random Forest parameters based on discovered research papers, and exploring MFCCs as TSI.

Appendices

Appendix A: Circuit Schematic (For Reference)



Appendix B: Code Snippets (Key Functions)

```
# Extracted from GUI/Features2.py
def extract_features(file_path):
    y, sr = librosa.load(file_path, sr=22050)
    mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=13)
    mfcc_mean = np.mean(mfcc.T, axis=0)
    mfcc_std = np.std(mfcc.T, axis=0)
    rmse = np.mean(librosa.feature.rms(y=y))
    zcr = np.mean(librosa.feature.zero_crossing_rate(y=y))
    # Returns 50-dimensional feature array for Random Forest Classifier
    return np.hstack((mfcc_mean, mfcc_std, rmse, zcr, ...))
```

Appendix C & D: Hardware Schematic & PCB Layout

Refer to the Github Repository to see the schematics of the PCB project.

Appendix E: Bill of Materials

- 1x STM32F401RCT6 BlackPill Development Board
- 1x MQ-5 Gas/Smoke Sensor Module
- 1x PIR Motion Sensor Module (HC-SR501)
- 1x NTC Thermistor Module
- 1x Photoresistor Module (LDR)
- 1x 5V DC Motor (F130)
- 1x SG90 Micro Servo Motor
- 1x Piezo Buzzer
- 1x RGB LED Module & 1x LED
- Miscellaneous Resistors, Capacitors.